VITALIY PIKALO

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SUMMARY

Computer Engineering student (B.S., Dec 2027) who ships production software in Python, C++, and JavaScript. Trained and deployed a 6.7M-parameter PyTorch reinforcement learning engine, built a fault-tolerant 24/7 pipeline handling 270K+ records, and had two pull requests merged into open-source ML libraries. Seeking a Summer 2027 SWE internship.

EDUCATION

The Pennsylvania State University University Park, PA
Bachelor of Science, Computer Engineering | GPA: 3.25 Expected Dec 2027

Relevant Coursework: Data Structures and Algorithms, Object-Oriented Programming, Statistics, Linear Algebra, Differential Equations, Calculus I-II, Digital Systems and Design

SKILLS

Languages: Python, C++, Java, JavaScript, Dart, SQL
Machine Learning and AI: PyTorch, TensorFlow, scikit-learn, NumPy, pandas, OpenCV, reinforcement learning, Monte Carlo Tree Search, LLM API integration
Backend and Data: FastAPI, REST API design, async Python, Node.js, Express, React, Flutter, SQLite, SQLAlchemy, MongoDB, Firebase, Supabase, Stripe API, concurrency control
Tools and Practices: Docker, Git, GitHub, code review, pytest, unit and regression testing, A/B experimentation, Linux, CUDA, Netlify
Certifications and Languages: Google AI (Google); Trading Comps Modeling (Wall Street Prep). Ukrainian (native), Russian (native), Spanish (intermediate)

EXPERIENCE

Full-Stack Software Engineer and Team Lead Jan 2025 - Present
Independent Contractor (Self-Employed) | *Erie, PA*

- Led a 4-engineer team to design, build, and ship 10+ production full-stack web and mobile applications serving 100+ customer reservations per month.
- Built secure payment and authentication services with the Stripe API and Supabase Auth, enforcing PCI-compliant transactions and role-based access control.
- Owned Git branching strategy, code review, and release workflow on GitHub across concurrent client projects.

Equity Analyst Aug 2025 - May 2026
Intrieri Family Student Managed Fund, Penn State Behrend | *Erie, PA*

- Co-managed a $1.6M live equity portfolio with a team of analysts under CFA Institute professional and ethical standards.
- Produced macroeconomic, industry, and valuation analyses that drove the fund's buy and sell decisions.

Artificial Intelligence Teaching Assistant and Guest Lecturer Jan 2026 - May 2026
The Pennsylvania State University | *Erie, PA*

- Delivered faculty-invited guest lectures to 100+ students on prompt engineering, AI-assisted developer workflows, and automation with Microsoft Copilot.
- Authored instructional materials and ran 1:1 technical sessions, measurably improving student assignment quality.

PROJECTS AND OPEN SOURCE CONTRIBUTIONS

Chess960-NN: AlphaZero-Style Reinforcement Learning Engine 2026
PyTorch, Python, FastAPI, Monte Carlo Tree Search, Reinforcement Learning, Hugging Face Spaces | *chess960-nn.netlify.app* | *github.com/Vitaliy-Pikalo/chess960-nn*

- Trained and deployed an AlphaZero-style Chess960 engine: a 10-block PyTorch ResNet (6.7M parameters) with PUCT Monte Carlo Tree Search.
- Served the model in production behind a FastAPI REST backend on Hugging Face Spaces; benchmarked to ~1600 Elo over 100 games vs Stockfish.

Kalshi Prediction-Market Trading Bot 2026
Python, SQLite, SQLAlchemy, httpx, NumPy, pandas, pytest, Kelly criterion, daemon architecture | *github.com/Vitaliy-Pikalo/kalshi-trading-bot*

- Engineered a fault-tolerant 24/7 daemon ingesting 270K+ live market snapshots into SQLite, surviving restarts with the database as single source of truth.
- Fixed concurrent-write failures via WAL journaling with a tuned busy-timeout and traced a ~18x data-inflation bug to an incorrect SQL join.

Open Source Contributions: lancedb/lancedb and marimo-team/marimo 2026
Python, PyTorch, JAX, vector databases, pytest | *Both reviewed by maintainers and merged into main*

- lancedb (PR #3567, merged): fixed a vector-column type-inference bug surfacing as an opaque runtime error; shipped an explicit ValueError plus a regression test.
- marimo (PR #9943, merged): fixed mo.audio to accept array-protocol objects such as torch.Tensor and JAX arrays, closing a gap with major ML frameworks.

LEADERSHIP, HONORS, AND ACTIVITIES

- **Varsity Men's Tennis, NCAA Division III:** AMCC Player of the Week and All-Tournament Team; 17-0 in singles; led the team to a 7-0 season, a 5th straight AMCC title, and an NCAA berth.
- **Leadership and Honors:** Student Government Senator, co-founded the campus AI Taskforce. Chess and Robotics Clubs: ran sponsored tournaments, built an autonomous rover. Dean's List, Spring 2026.